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Nonlinear State Estimation Based on Continuous-Discrete Cubature Kalman Filter for Fermentation Process

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Abstract: To realize on-line detection of biological parameters in fermentation process, a method of nonlinear state estimation for fermentation process based on continuous-discrete Cubature Kalman Filter (CD-CKF) is proposed. The nonlinear state estimation is realized by analyzing the model of fermentation process and applying the CD-CKF which takes use of Gaussian approximation filter approach to update system state and covariance. Then, the implementation steps of the algorithm are given. The experimental results show that the proposed method has high estimation accuracy when dealing with the on-line estimation of fermentation process.

Key Words: nonlinear state estimation, continuous-discrete CKF, fermentation process

1 Introduction

The fermentation process is a highly variable, nonlinear and complex process with quantities of variable. Some critical biomass parameters need off-line analysis, and it is difficult to measure them on-line [1]. In addition, the sampling operation is easy to cause bacterial infection for fermentation process [2]. Therefore, nonlinear Gauss filter is an effective method for on-line detection of critical biomass parameters in nonlinear state estimation. Describing the dependency relationship of specific quantity attributes of time and space is a key point in modeling fermentation processes. Using differential equations is a generic strategy to describe the change of state variables [3]. Because measurements are often based on fixed sampling interval detection, the measurement equations are often discrete for fermentation process，while the system equations of the fermentation process model are mostly continuous. Therefore, this nonlinear state-space model for the fermentation process can be taken as a continuous-discrete model.

Previously, in order to solve the state estimation problem for the continuous-discrete model, continuous state equations were discretized, and then nonlinear Kalman filtering or other state estimation methods were used to solve the problem. Bellgardt et al. [4] used extended Kalman filter (EKF) to realize state estimation of some key parameters in the fermentation process. This method is simple, but when the nonlinear state equations have strong non-linearity or lots of parameters, the process of discretization will be very complicated or even infeasible. Haseltine et al. [5] pointed out that in their experiments, a large number of EKF failures had occurred and led to a large amount of negative effects on subsequent experiments. In order to solve the obstacles in the discretization process, a method which deals with the state estimation of continuous-discrete systems directly is presented. At first, the continuous-discrete method was applied to extended Kalman filter (EKF), and the state estimation of the model was preferably solved [6-7]. However, there are many weaknesses in EKF itself such as filtering divergence and low precision in some systems. Later, Särkkä et al. [8] applied continuous-discrete method to unscented Kalman filter (UKF) which played an important role in dealing with strong nonlinear systems, and achieved good results. Xu et al. [9] improved continuous-discrete UKF to make covariance matrix positive definite and avoided the square root being imaginary. Li et al. [10] compared continuous- discrete UKF and continuous-discrete EKF on strongly nonlinear systems, and proved the advantages of continuous-discrete UKF. However, the UKF method has a problem in the selection of parameters, and the covariance matrix may be negative in the calculation process so that it cannot be calculated.

Recently, Arasaratnam et al. [11] proposed the cubature Kalman filter (CKF). Compared with the UKF, this method has a remarkable performance, better stability and less calculation in the filter processing [12]. Therefore, the CKF is more suitable when dealing with the nonlinear state estimation of the strong nonlinear system with high order. Särkkä et al. [13] also proposed the continuous-discrete CKF (CD-CKF) based on the CKF method. The continuous-discrete CKF modifies the iterative method of the covariance matrix in the calculation of sigma points. This method is supposed to have better accuracy and stability when it is applied to the nonlinear state estimation of fermentation process than continuous-discrete EKF and continuous-discrete UKF. However, the CD-CKF has only been utilized in motion systems so far. Other literatures have not reported the use of continuous-discrete CKF methods for the nonlinear state estimation in fermentation processes.

In this paper, we apply the continuous-discrete CKF to state estimation of fermentation processes and realize the online measurement of key biological parameters. The rest of the paper is organized as follows. In section 2, the general model of fermentation process is analyzed. The nonlinear state estimation method based on the CD-CKF is discussed in section 3. The flow chart of the CD-CKF is also given in section 3. In section 4 the proposed method is verified by the simulation experiments of fermentation process. Concluding summary is in section 5.

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2 Continuous-Discrete Model of Fermentation Process

Fermentation is a very complex biochemical process involving intracellular biological responses, intracellular and extracellular substances, the transmission and response of extracellular substances [3]. The parameters of the process are numerous. Generally speaking, state parameters and biological parameters are two kinds of parameters in conventional fermentation systems. Most of the state parameters can be measured online by hardware sensors, while some of the key biological parameters need to be analyzed by laboratory using special instruments, and the measurement has a long delay, which decreases the accuracy. The state estimation using the nonlinear filter is a preferably way to solve such problems.

The fermentation process mainly includes batch fermentation, fed batch fermentation and continuous fed batch fermentation. Among these, fed batch fermentation is the most common type. In fed batch fermentation, the equation of conservation of material can be expressed by formula (1) [14]

$$\frac{d\boldsymbol{\xi}}{dt} = \boldsymbol{K}_m \boldsymbol{r}(\boldsymbol{\xi}) + \boldsymbol{u} \tag{1}$$

where $\boldsymbol{\xi}$ represents the state vector of the concentration of specific components in the fermentation process. $\boldsymbol{K}_m \boldsymbol{r}(\boldsymbol{\xi})$ represents the changes of biochemical and biological reactions during fermentation. $\boldsymbol{r}(\boldsymbol{\xi})$ represents the reaction rate vector of different components. $\boldsymbol{K}_m$ represents the chemical calculation matrix. $\boldsymbol{u}$ represents input vector in fermentation tank.

When the measurable variables are added to the fermentation process, the state space equation of the system is shown like formula (2)

$$\begin{cases} \dfrac{\mathrm{d}\boldsymbol{x}(t)}{\mathrm{d}t} = f(\boldsymbol{x}(t), \boldsymbol{u}(t)) + \boldsymbol{w}(t) \\ \boldsymbol{z}(t) = h(\boldsymbol{x}(t)) + \boldsymbol{v}(t) \end{cases} \tag{2}$$

where $\boldsymbol{x}$ is the state vector. $\boldsymbol{z}$ is the measurement vector. $\boldsymbol{u}$ is the input vector. $\boldsymbol{w}$ and $\boldsymbol{v}$ respectively represent the Gauss white noise vector whose mean are zero and the variance respectively are $\boldsymbol{Q}$ and $\boldsymbol{R}$. f and h are respectively linear or nonlinear functions of state transformation and measurement output.

For the formula (2), if the nonlinear Kalman filter is used directly, the discretization problem will be encountered when calculating the sampling point. There are two ways dealing with the problem. One is discretizing the state equation first, and the other one is discretizing the system when filtering. It is an intuitive method to discretize the state equation first, and then filtering it, but sometimes the discretization process is very complicated, or even infeasible. Discretizing the system when filtering can relatively avoid the problem of tedious calculation.

The earliest continuous discrete method was applied to EKF and continuous-discrete EKF was proposed. The main innovation of the method is proposing a new method for solving the covariance matrix $\boldsymbol{P}$, which treats each element of the $\boldsymbol{P}$ matrix as a function and then integrate. However, due to the limitation of the EKF itself, the method needs complex operations such as derivation, and the filtering effect is not satisfactory in the face of high nonlinearity or complex state equations.

3 Nonlinear State Estimation Based on the CD-CKF

By combining the continuous-discrete method with CKF, difficult calculation can be avoided when taking the derivative of complex equation and the problem that EKF exists tremendous limitations when dealing with high-order nonlinear systems. The measurement update of continuous-discrete CKF method is consistent with CKF method, while the time updating is also consistent with CKF in addition to the updating of the covariance matrix $\boldsymbol{P}$.

3.1 CKF algorithm

The CKF transforms the integral form in the state space into the integral form under polar coordinates, and uses the cubature rule to do the numerical integration of the Bayesian posterior probability density function. The concrete implementation steps are as follows.

Consider the following nonlinear discrete systems

$$\begin{cases} \boldsymbol{x}_k = \boldsymbol{f}(\boldsymbol{x}_{k-1}, \boldsymbol{u}_{k-1}) + \boldsymbol{\omega}_{k-1} \\ \boldsymbol{z}_k = \boldsymbol{h}(\boldsymbol{x}_k) + \boldsymbol{\upsilon}_k \end{cases} \tag{3}$$

where $\boldsymbol{x}_k$ is the n-dimensional state vector of the system. $\boldsymbol{z}_k$ is the measurement vector of the system. $\boldsymbol{f}_k(\cdot)$ is the state function of the system. $\boldsymbol{h}_k(\cdot)$ is the measurement function of the system. $\boldsymbol{\omega}_k$ is the system noise. $\boldsymbol{\upsilon}_k$ is the measurement noise. $\boldsymbol{\omega}_k$ and $\boldsymbol{\upsilon}_k$ are independence. $\boldsymbol{\omega}_k \sim N(0, \boldsymbol{Q}_k)$, $\boldsymbol{\upsilon}_k \sim N(0, \boldsymbol{R}_k)$. $\boldsymbol{P}_{0|0}$ is the initial value of the error covariance matrix. $\boldsymbol{S}_{0|0} = chol(\boldsymbol{P}_{0|0})^{\mathrm{T}}$. The specific steps of the CKF are shown as follows.

Time update

1. Update the cubature point by $\boldsymbol{S}_{k-1|k-1}$ at the previous moment

$$\boldsymbol{X}_{i,k-1|k-1} = \boldsymbol{S}_{k-1|k-1}\boldsymbol{\xi}_i + \hat{\boldsymbol{x}}_{k-1|k-1} \tag{4}$$

$$\boldsymbol{X}^*_{i,k|k-1} = \boldsymbol{f}(\boldsymbol{X}_{i,k-1|k-1}, \boldsymbol{u}_{k-1}) \tag{5}$$

2. Update status prediction for one step

$$\hat{\boldsymbol{x}}_{k|k-1} = \frac{1}{m}\sum_{i=1}^{m} \boldsymbol{X}^*_{i,k|k-1} \tag{6}$$

3. Update prediction error covariance for one step

$$\boldsymbol{P}_{k|k-1} = \frac{1}{m}\sum_{i=1}^{m} \boldsymbol{X}^*_{i,k|k-1}\boldsymbol{X}^{*\mathrm{T}}_{i,k|k-1} - \hat{\boldsymbol{x}}_{k|k-1}\hat{\boldsymbol{x}}^{\mathrm{T}}_{k|k-1} + \boldsymbol{Q}_{k-1} \tag{7}$$

Measurement update

1. Calculate the cubature point

$$\boldsymbol{S}_{k|k-1} = \mathrm{chol}(\boldsymbol{P}_{k|k-1})^{\mathrm{T}} \tag{8}$$

$$\boldsymbol{X}_{i,k|k-1} = \boldsymbol{S}_{k|k-1}\boldsymbol{\xi}_i + \hat{\boldsymbol{x}}_{k|k-1} \tag{9}$$

2. Update the measurement equations and propagate the cubature points

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$$\boldsymbol{Z}_{i,k|k-1}=\boldsymbol{h}\left(\boldsymbol{X}_{i,k|k-1}\right) \tag{10}$$

3. Update predicted value of the measurement

$$\hat{\boldsymbol{z}}_{k|k-1}=\frac{1}{m}\sum_{i=1}^{m}\boldsymbol{Z}_{i,k|k-1} \tag{11}$$

4. Update predicted output covariance

$$\boldsymbol{P}_{zz,k|k-1}=\frac{1}{m}\sum_{i=1}^{m}\boldsymbol{Z}_{i,k|k-1}\boldsymbol{Z}_{i,k|k-1}^{\mathrm{T}}-\hat{\boldsymbol{z}}_{k|k-1}\hat{\boldsymbol{z}}_{k|k-1}^{\mathrm{T}}+\boldsymbol{R}_{k} \tag{12}$$

5. Update the cross covariance

$$\boldsymbol{P}_{xz,k|k-1}=\frac{1}{m}\sum_{i=1}^{m}\boldsymbol{X}_{i,k|k-1}\boldsymbol{Z}_{i,k|k-1}^{\mathrm{T}}-\hat{\boldsymbol{x}}_{k|k-1}\hat{\boldsymbol{z}}_{k|k-1}^{\mathrm{T}} \tag{13}$$

6. Update the Kalman filter gain

$$\boldsymbol{W}_{k}=\boldsymbol{P}_{xz,k|k-1}\boldsymbol{P}_{zz,k|k-1}^{-1} \tag{14}$$

7. Update the predicted status values

$$\hat{\boldsymbol{x}}_{k|k}=\hat{\boldsymbol{x}}_{k|k-1}+\boldsymbol{W}_{k}\left(\boldsymbol{z}_{k}-\hat{\boldsymbol{z}}_{k|k-1}\right) \tag{15}$$

8. Updating the error covariance matrix and its characteristic square root

$$\boldsymbol{P}_{k|k}=\boldsymbol{P}_{k|k-1}-\boldsymbol{W}_{k}\boldsymbol{P}_{zz,k|k-1}\boldsymbol{W}_{k}^{\mathrm{T}} \tag{16}$$

$$\boldsymbol{S}_{k|k}=\mathrm{chol}\left(\boldsymbol{P}_{k|k}\right)^{\mathrm{T}} \tag{17}$$

where the set of cubature points is given by

$$\xi_{i}=\begin{cases}\sqrt{n}\boldsymbol{e}_{i},i=1,2,...,n\\-\sqrt{n}\boldsymbol{e}_{i-n},i=n+1,...,2n\end{cases} \tag{18}$$

where $\boldsymbol{e}_{i}$ denotes the ith column of a n-dimensional unit matrix; m represents the number of cubature points. For three-order cubature criterion, the total number of cubature points is 2 times the dimension of the state. That is, m=2n.

3.2 Continuous-discrete CKF

The difference between the continuous-discrete CKF and CKF is that the solution form of the covariance matrix $\boldsymbol{P}$ is different during time updating. The concrete deduction process is as follows. Based on the Kalman filter, formulas (19-20) can be achieved.

$$\frac{\mathrm{d}\boldsymbol{x}(t)}{\mathrm{d}t}=E[(f(\boldsymbol{x},t)] \tag{19}$$

$$\frac{\mathrm{d}\boldsymbol{P}}{\mathrm{d}t}=E[(\boldsymbol{x}-\boldsymbol{x}_{0})f^{\mathrm{T}}(\boldsymbol{x},t)]+E[f(\boldsymbol{x},t)(\boldsymbol{x}-\boldsymbol{x}_{0})^{\mathrm{T}}]+\boldsymbol{Q} \tag{20}$$

By applying Gauss filter approximation method to formulas (19-20), the key equations of time update part of the CD-CKF can be achieved.

$$\frac{\mathrm{d}\boldsymbol{x}(t)}{\mathrm{d}t}=\frac{1}{2n}\sum_{i=1}^{2n}f(\boldsymbol{x}(t)+\sqrt{\boldsymbol{P}(t)}\xi_{i},t) \tag{21}$$

$$\begin{aligned}\frac{\mathrm{d}\boldsymbol{P}(t)}{\mathrm{d}t}&=\frac{1}{2n}\sum_{i=1}^{2n}f(\boldsymbol{x}(t)+\sqrt{\boldsymbol{P}(t)}\xi_{i},t)\xi_{i}^{\mathrm{T}}\sqrt{\boldsymbol{P}(t)}^{\mathrm{T}}\\&+\frac{1}{2n}\sum_{i=1}^{2n}\sqrt{\boldsymbol{P}(t)}\xi_{i}f^{\mathrm{T}}(\boldsymbol{x}(t)+\sqrt{\boldsymbol{P}(t)}\xi_{i},t)+\boldsymbol{Q}\end{aligned} \tag{22}$$

The priori estimation of $\boldsymbol{X}$ and $\boldsymbol{P}$ at k moment can be computed by integrating d$\boldsymbol{X}$ and d$\boldsymbol{P}$ from k-1 moment to k moment. In particular, when integrating covariance matrix $\boldsymbol{P}$, each element in the covariance matrix $\boldsymbol{P}$ is treated as an equation and integrate from k-1 moment to k moment like continuous-discrete EKF. In this paper, the integral method and the specific implementation steps of the CD-CKF are given explicitly. In the actual process, Euler numerical is used to integrate each equation. Therefore, the change of the CD-CKF is only the time update procedure in Section 3.1, while the other parts are exactly the same as those in the 3.1 section.

3.3 Implementation of the continuous-discrete CKF algorithm

This paper realizes the nonlinear state estimation of fermentation system based on the continuous-discrete CKF. The implementation flow of the CD-CKF algorithm is shown in Fig. 1. The concrete steps are shown as follows.

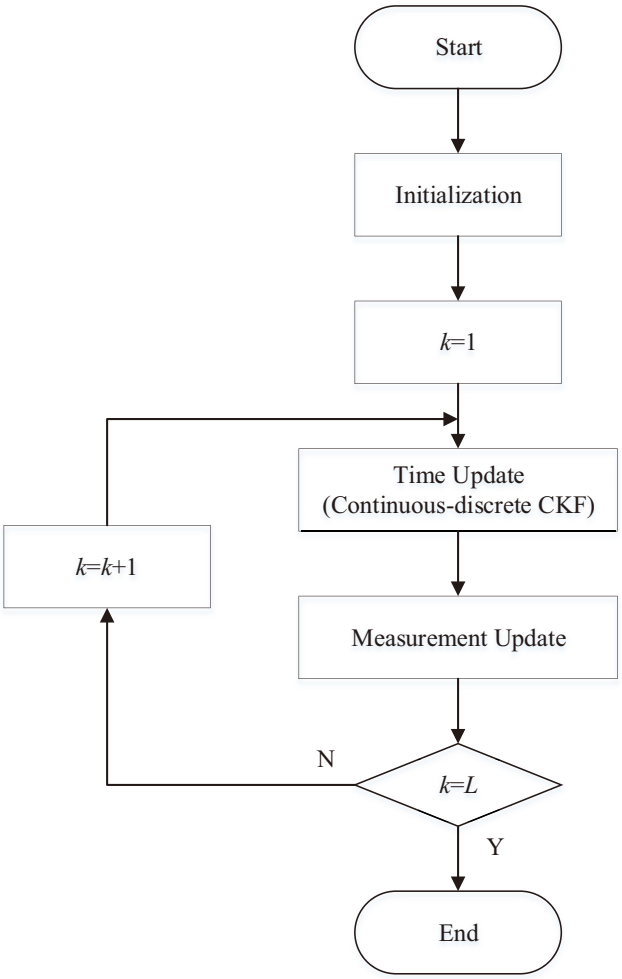


Fig. 1: Flow chart of the CD-CKF

1. Set the initial value of state, the initial value of error covariance, the variance of system noise and the variance of measurement noise first. The time series are $k\in\{1,\cdots,L\}$.
2. According to state equation, the differential form of measurement $\boldsymbol{x}$ and covariance matrix $\boldsymbol{P}$ in formulas (21-22) can be calculated, and then integrate $\boldsymbol{x}$ and $\boldsymbol{P}$. It is important to note that the integral of the matrix $\boldsymbol{P}$ is to use each item of the $\boldsymbol{P}$ matrix as a variable to solve the integral, while the other quantities are constants at this time.
3. Measurement update part is consistent with the CKF method in section 3.1.

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4. Determine whether k is equal to L, and if not, k=k+1, then go to the second step and continue the cycle.

4 Simulation Experiments

The classical three-order fermentation model used by Chitralekha et al. [15] is adopted in this paper. The state equation is shown in formulas (23-26), and the measurement equation is shown in formula (27).

$$\frac{\mathrm{d}X}{\mathrm{d}t} = -DX + \mu(P,S)X \tag{23}$$

$$\frac{\mathrm{d}S}{\mathrm{d}t} = D(S_f - S) - \frac{1}{Y_{X/S}}\mu(P,S)X \tag{24}$$

$$\frac{\mathrm{d}P}{\mathrm{d}t} = -DP + (\alpha\mu(P,S) + \beta)X \tag{25}$$

$$\mu(P,S) = \frac{\mu_m(1-(P/P_m))S}{K_m + S + (S^2/K_i)} \tag{26}$$

$$z = [X,P] \tag{27}$$

X refers to biomass concentration, S refers to substrate concentration and P refers to product concentration. D refers to dilution rate. S_f refers to feed substrate concentration. $Y_{X/S}$, α and β are all yield parameters. u_m refers to maximum specific growth. P_m, K_m, K_i are reaction constants. $u(P,S)$ refers to Specific growth rate. X and P are observables, the observation equation is linear, and S is the estimator. D and S_f are a set of related values, and when D is between 0.1h^{-1} and 0.3h^{-1}, the corresponding S_f is between 10g/L and 30g/L. The remaining values are constant, and the values and units of the above parameters are shown in Table 1.

Table 1: The parameters of fermentation process

Parameters	Values
μ_m	0.48h^{-1}
K_i	1.2g/L
$1/P_m$	0.02L/g
$1/K_m$	0.0455L/g
$1/Y_{x/s}$	2.5g/g
α	2.2g/g
β	0.2 h-1

The noise in the experiment is Gauss white noise whose mean is 0, the system noise and measurement noise are given as follows:

The system noise: $\boldsymbol{Q}$=[0.01,0,0;0,0.1,0;0,0,0.01]

The measurement noise: $\boldsymbol{R}$=[0.1,0;0,0.1]

The initial values of true value, state estimation and state covariance are respectively given as follows.

The initial values of true value: $\boldsymbol{X}(0)=[1,5,1]^{\mathrm{T}}$

The initial values of state estimation: $\boldsymbol{X}'(0)=[2\ 6\ 2]^{\mathrm{T}}$

The initial values of state covariance: $\boldsymbol{P}(0)$=diag [0.01 ,0.1, 0.01]

The sampling interval is 1h. The CD-CKF and EKF with the Euler method are used to estimate the fermentation states. Euler discretization step length is 1h. Since both $\boldsymbol{X}$ and $\boldsymbol{P}$ are the measurements in the experiments, the estimation performance is better, and the estimation errors are mainly existed in $\boldsymbol{S}$. The results of the experimental simulation are shown in Figs. 2-4.

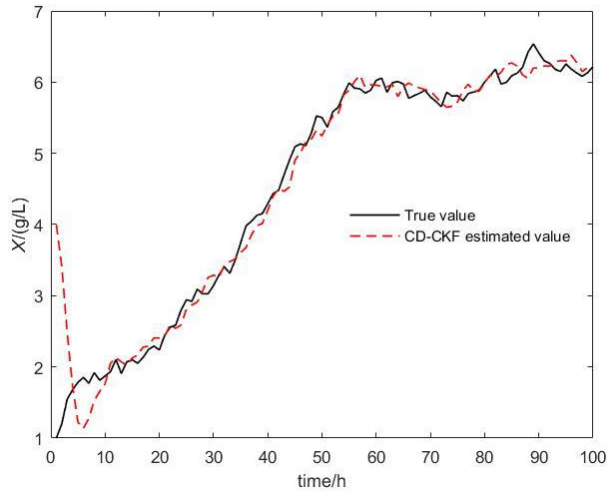


Fig. 2: Estimation of the biomass concentration

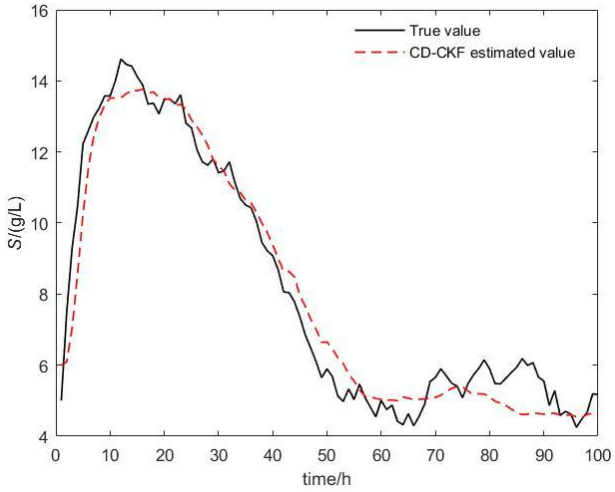


Fig. 3: Estimation of the substrate concentration

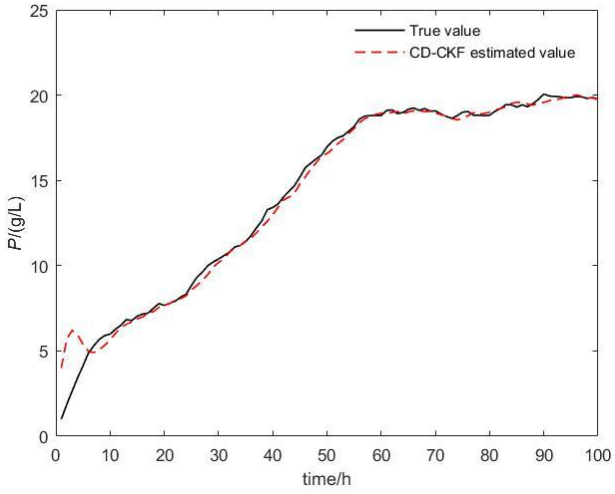


Fig. 4: Estimation of the product concentration

We take two indexes, namely absolute error and relative error to evaluate the accuracy. The absolute error uses MAPE (Mean Absolute Percent Error) and RMSPE(Root Mean Square Percentage Error). In this experiment, the

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evaluation accuracy of MAPE can be referenced to the following methods: less than or equal to 10%, the estimation accuracy is very high; 10% to 20%,the estimation accuracy is high; 20% to 50%, the estimation accuracy is acceptable; 50% above, the estimation accuracy is very low. Because of the relation between the square terms, RMSPE is easy to be affected by outliers, and gives greater weight to the larger errors, which leads to larger results, but RMSPE can still judge the pros and cons of the model like MAPE [16].

Since the noise in this experiment is random, it needs repeated experiments to get a accurate evaluation. Therefore, the average MAPE and RMSPE of the 50 experimental results are used to evaluate the estimation accuracy. As shown in Table 2, the estimation errors of each measurement are less than 10% using MAPE method and we consider that the estimation accuracy is very high. Although the relative error of RMSPE method is relatively large due to the method itself, but it is still within the acceptable range.

Table 2: The relative error of each measurement

	MAPE	RMSPE
Relative error of biomass concentration	6.00%	13.66%
Relative error of substrate concentration	7.33%	9.51%
Relative error of product concentration	4.78%	14.06%

The relative errors are compared by RMSE (Root Mean Square Error) and ARMSE (Average RMSE). When the initial values and parameters are fixed, compare RMSE and ARMSE of the CD-CKF and EKF in different simulation times. As shown in Figs. 5-7 and Table 3, RMSE and ARMSE of the CD-CKF are all less than EKF for the estimation errors of state quantities $\boldsymbol{X}$, $\boldsymbol{S}$, and $\boldsymbol{P}$.

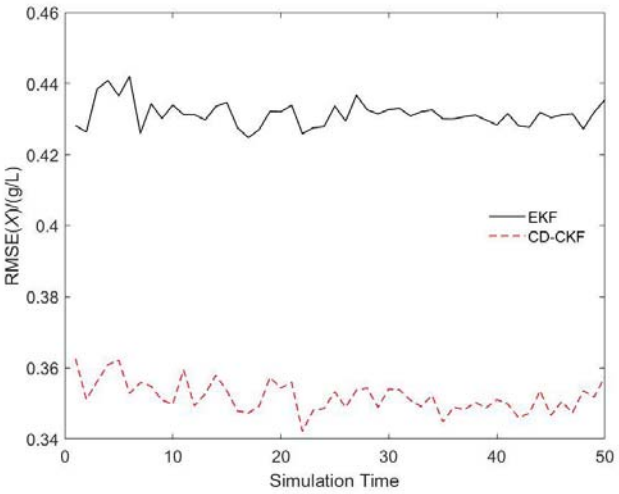


Fig. 5: RMSE of the biomass concentration

Table 3: The contrast of ARMSE between CD-CKF and EKF

	EKF(g/L)	CD-CKF(g/L)
ARMSE of biomass concentration	1.10	0.89
ARMSE of substrate concentration	2.00	1.68
ARMSE of product concentration	2.35	1.89

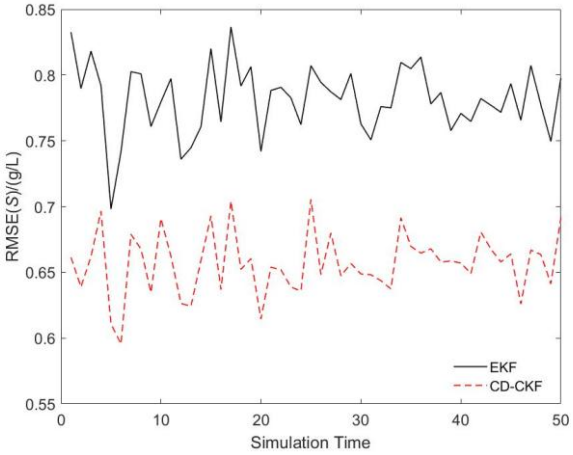


Fig. 6: RMSE of the substrate concentration

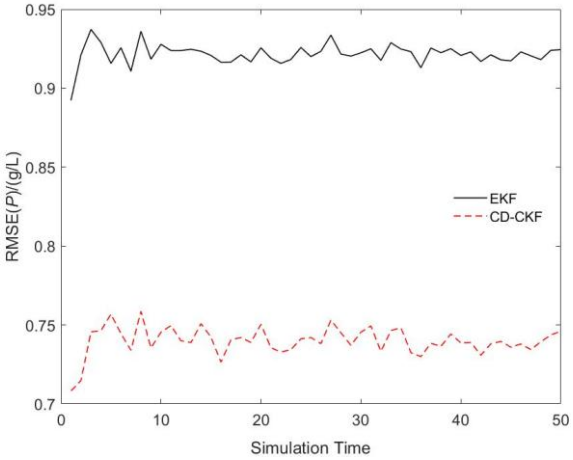


Fig. 7: RMSE of the product concentration

Using Matlab 2015b, the average single simulation time of 50 times is only 0.0246 seconds on the computer with CPU I7-6700hq, 2.6GHz, 16G DDR4 memory. The algorithm can meet the real-time requirements. Therefore, the CD-CKF is feasible for estimating the state of fermentation processes on-line.

5 Conclusions

This paper uses the continuous-discrete CKF to deal with the state estimation of fermentation systems. The method reduces the computational complexity in the discretization process by using the Gauss approximation first and then discretizing the state equations and the estimation accuracy is relatively high. The method has a good performance on the models with the continuous system model and the discrete measurement model and has a good application prospect in the nonlinear state estimation of fermentation processes.

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